

Monitor & Troubleshoot **Virtual Networking**

Network Watcher tools, NSG flow logs, connection monitors, diagnostics, and a systematic troubleshooting approach for every common Azure networking failure pattern.

The Network Watcher toolkit — pick the right tool

Network Watcher is a regional service (auto-enabled when a VNet is created) that groups all Azure networking diagnostic tools in one place. Each tool answers a different question — knowing which one to reach for first is what the exam tests.

FIRST STOP

IP Flow Verify

"Is this port blocked?"

Tests whether a specific packet is allowed or denied by NSG rules — and names the exact rule responsible.

ROUTING

Next Hop

"Where is traffic going?"

Shows the next hop type and IP for traffic from a VM — confirms UDRs, detects routing loops.

END-TO-END

Connection Troubleshoot

"Can A reach B on port X?"

Tests full connectivity between two endpoints — includes NSG, routing, and TCP handshake checks.

ONGOING

Connection Monitor

"Is connectivity stable over time?"

Continuously monitors connectivity between sources and destinations — alerts on latency or packet loss changes.

DEEP INSPECT

Packet Capture

"What packets are actually sent?"

Captures NIC-level packets to a storage account or local file — last resort for layer-by-layer inspection.

AUDIT

NSG Flow Logs

"What traffic has been flowing?"

Records all allowed and denied flows through an NSG with source IP, dest IP, port, protocol, and direction.

VISUAL

Topology

"What does my network look like?"

Visual map of all resources in a VNet — VMs, NICs, subnets, NSGs, peerings, and gateways.

VPN

VPN Troubleshoot

"Why is my VPN tunnel down?"

Diagnoses VPN Gateway and connection health — identifies BGP issues, tunnel state, and IKE errors.

Eight terms you need cold

Network Watcher

The regional Azure service that hosts all network monitoring and diagnostic tools. Must be enabled per region — auto-enabled when a VNet is created in that region.

Tip: Network Watcher lives in a resource group named `NetworkWatcherRG` — don't delete it or diagnostics will fail.

NSG Flow Logs

A Network Watcher feature that logs every flow through an NSG to a storage account — includes source/dest IP, ports, protocol, direction, and allow/deny decision.

Tip: Version 2 flow logs add bytes and packets per flow. Enable Traffic Analytics on top for aggregated visualizations in Log Analytics.

→ Traffic Analytics

A service that processes NSG flow logs and sends aggregated data to Log Analytics — provides dashboards for top talkers, malicious IPs, port utilization, and geo distribution.

Tip: Traffic Analytics adds cost (data processing + Log Analytics ingestion) but is far more useful than raw flow log JSON for operational insight.

⌘ Connection Monitor

Continuously tests connectivity between sources (VMs, agents) and destinations (endpoints, URLs) — tracks latency trends and alerts on failures over time.

Tip: Replaces the older Connection Monitor (classic) and Network Performance Monitor. Supports Azure and on-premises sources via Log Analytics agent.

📄 Packet Capture

Captures packets at the NIC level of a VM and saves them to a storage account or a local file — uses the Network Watcher VM extension on the target VM.

Tip: Requires the Azure Network Watcher Agent VM extension to be installed on the target VM — install it before you need it, not during an outage.

✓ Effective Security Rules

A portal view (VM → Networking → Effective security rules) that merges subnet NSG and NIC NSG rules in priority order — the fastest way to see the actual combined rules applying to a VM.

Tip: Use this before running IP Flow Verify — it shows all rules at once rather than testing one packet at a time.

> Effective Routes

Shows all routes applying to a NIC — system defaults, VNet peering routes, and UDRs merged together in the order they're evaluated. Available in portal or CLI.

Tip: Use Effective Routes to verify that a UDR is actually being applied — if a custom route doesn't appear here, check the route table association on the subnet.



Azure Monitor — Network Insights

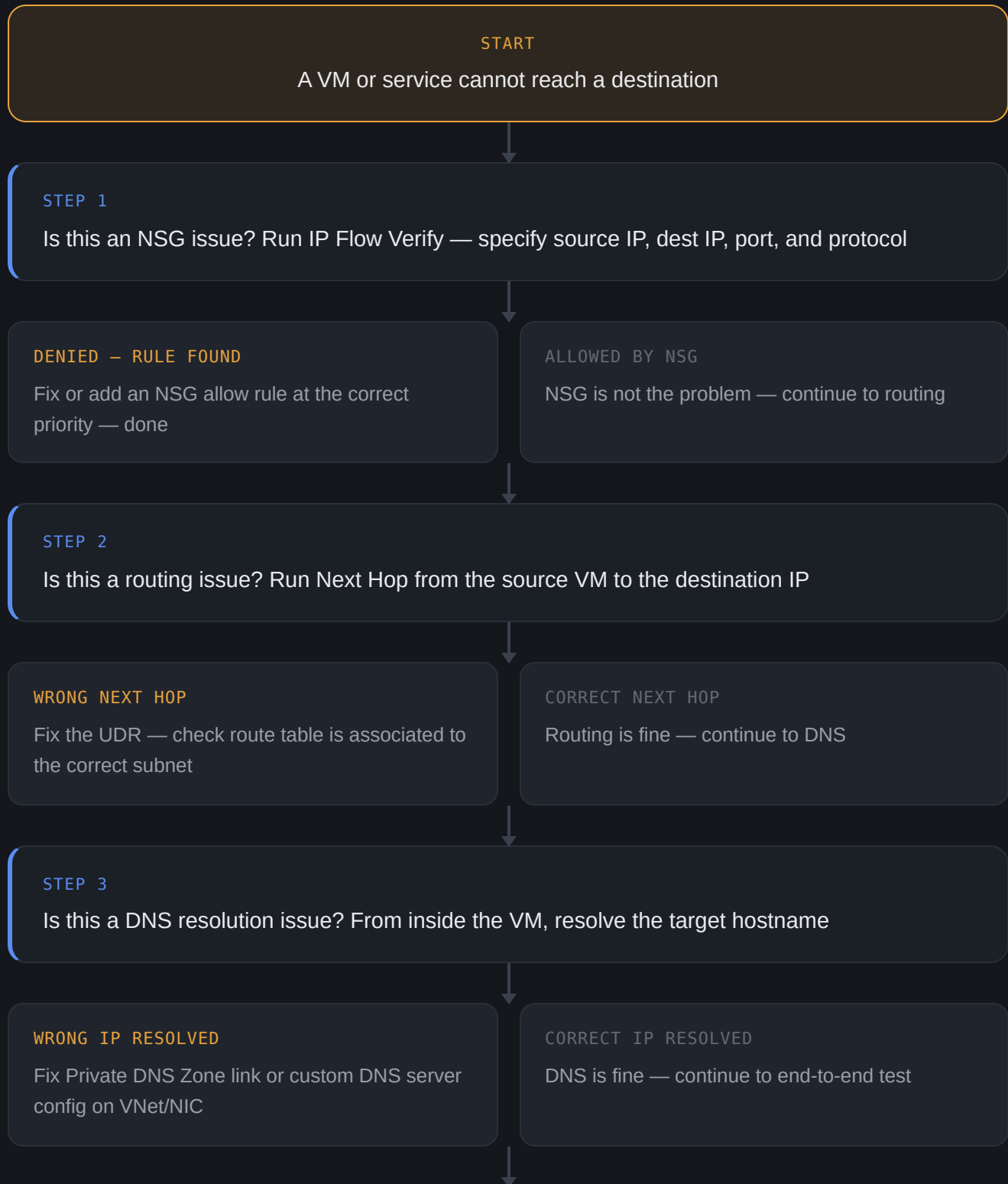
A unified dashboard in Azure Monitor that aggregates health, metrics, and alerts across all networking resources — Load Balancers, Application Gateways, VPN Gateways, ExpressRoute, and VNets.

Tip: Network Insights uses workbooks — you can customize the views and pin them to dashboards for team-wide visibility.

TOOL	REAL-TIME?	HISTORICAL?	REQUIRES VM AGENT?	BEST EXAM GOTCHA — NETWORK WATCHER AGENT VS. NO AGENT FIRST STEP FOR	
IP Flow Verify	Yes	No	No	IP Flow Verify and Next Hop work with no agent — they query the Azure control plane directly. Connection Troubleshoot and Packet Capture require the Network Watcher Agent VM extension on the target VM. If you run Connection Troubleshoot and get an error about the extension, install it first: <code>AzureNetworkWatcherExtension</code> on Windows, <code>NetworkWatcherAgentLinux</code> on Linux.	End-to-end TCP connectivity Ongoing latency / availability Audit, forensics, unexplained drops Deep packet inspection
Next Hop	Yes	No	No	Routing / UDP not working	
Connection Troubleshoot	Yes	No	Yes		
Connection Monitor	Yes	Yes	Yes		
NSG Flow Logs	No (async)	Yes	No		
Packet Capture	Yes	Yes	Yes		

Systematic network troubleshooting flow

Follow this order every time — it eliminates the most common causes first without wasting time on deep inspection tools when a simpler fix exists.



STEP 4

Run Connection Troubleshoot between the source VM and destination — does it succeed?

FAILS — HOPS SHOWN

Identify which hop fails — check NSG/firewall at that hop

SUCCEEDS

Network is healthy — issue is at the application layer

STEP 5

Is this an intermittent or historical issue?

INTERMITTENT

Enable NSG Flow Logs + Traffic Analytics — review historical traffic patterns

ONGOING / DEEP

Run Packet Capture to inspect actual packets at the NIC level

DONE

Root cause identified at NSG, routing, DNS, or application layer

Network Watcher CLI commands

>_ IP Flow Verify & Next Hop

BEST FOR: NSG and routing diagnosis – always start here

No agent required — runs immediately against the control plane. Fastest first step for any connectivity issue.

```
# IP Flow Verify – test if port 443
is allowed inbound
az network watcher test-ip-flow \
  --resource-group rg-prod \
  --vm vm-web01 \
  --direction Inbound \
  --protocol TCP \
  --local 10.0.1.4:443 \
  --remote 203.0.113.50:54321
```

```
# Next Hop – where does traffic to
10.0.3.10 go?
az network watcher show-next-hop \
  --resource-group rg-prod \
  --vm vm-web01 \
  --source-ip 10.0.1.4 \
  --dest-ip 10.0.3.10
```

>_ Connection Troubleshoot & Monitor

BEST FOR: end-to-end connectivity tests and ongoing monitoring

Tests full TCP path including NSG, routing, and connectivity. Connection Monitor sets up continuous tests that alert on failures.

```
# One-time connection test
az network watcher test-
connectivity \
  --resource-group rg-prod \
  --source-resource vm-web01 \
  --dest-address 10.0.3.10 \
  --dest-port 1433
```

```
# Create a Connection Monitor
az network watcher connection-
monitor create \
  --name mon-web-to-sql \
  --resource-group rg-prod \
  --location eastus \
  --source-resource vm-web01 \
  --dest-address 10.0.3.10 \
  --dest-port 1433 \
  --monitoring-interval 30
```


>_ NSG Flow Logs & Traffic Analytics

BEST FOR: auditing, forensics, traffic analysis

Enable NSG Flow Logs to capture all traffic through an NSG. Add Traffic Analytics to get aggregated dashboards in Log Analytics.

```
# Enable NSG flow logs (v2) az
network watcher flow-log create \
--resource-group rg-prod \ --nsg
nsg-web \ --storage-account
mystorageacct \ --enabled true \ --
format JSON \ --log-version 2 \ --
retention 30 # Enable Traffic
Analytics az network watcher
flow-log update \ --resource-group
rg-prod \ --nsg nsg-web \ --
workspace
/subscriptions/.../workspaces/law-
prod \ --traffic-analytics true \ --
interval 10
```

>_ Effective Rules & Routes

BEST FOR: viewing merged NSG rules and routes on a NIC

See all effective NSG rules and routes for a specific NIC — combines subnet and NIC-level rules into one merged view.

```
# Effective NSG security rules on a
NIC
az network nic list-effective-nsg \
--resource-group rg-prod \
--name nic-vm-web01 \
--output table

# Effective routes on a NIC
az network nic show-effective-
route-table \
--resource-group rg-prod \
--name nic-vm-web01 \
--output table
```

>_ Packet Capture

BEST FOR: deep packet-level inspection
(last resort)

Captures packets from a VM NIC to storage
— requires Network Watcher Agent extension.
Use for issues that survive all higher-level
checks.

```
# Start a packet capture (requires
NW agent on VM)
az network watcher packet-capture
create \
  --resource-group rg-prod \
  --vm vm-web01 \
  --name capture-web01 \
  --storage-account mystorageacct \
  --time-limit 300 \
  --filters
'[{ "protocol": "TCP", "localPort": "80"
}]'

# Stop and download
az network watcher packet-capture
stop \
  --resource-group rg-prod \
  --name capture-web01 \
  --location eastus
```

SCENARIO	FIRST TOOL	SECOND TOOL IF FIRST PASSES
VM can't receive traffic on a port	IP Flow Verify	Effective Security Rules
Traffic not going through firewall/NVA	Next Hop	Effective Routes
Can't resolve private DNS name	nslookup inside VM	Check Private DNS Zone VNet link
Intermittent connection drops	Connection Monitor	NSG Flow Logs + Traffic Analytics
VPN tunnel down	VPN Troubleshoot	Azure Monitor VPN Gateway metrics
Unknown packet behavior	Connection Troubleshoot	Packet Capture

Principles worth internalizing

TRAFFIC ANALYTICS – WORTH ENABLING FOR PRODUCTION

Raw NSG Flow Logs are JSON blobs in a storage account — useful but hard to query. **Traffic Analytics** processes those logs and sends aggregated data to **Log Analytics**, giving you prebuilt workbooks for top talkers, anomalous flows, blocked ports, and geo distribution. Enable it on every production NSG with a 10-minute aggregation interval. The additional cost is usually justified by the operational visibility it provides.

EXAM GOTCHA – DELETING NETWORKWATCHERRG

Network Watcher resources live in a resource group called `NetworkWatcherRG` created automatically per region. **Deleting this resource group disables all Network Watcher tools for that region** — including packet captures already running and connection monitors. Never delete NetworkWatcherRG in production. The exam sometimes presents it as something to clean up to save cost — it is not.

Six mistakes that show up on the exam

Using Packet Capture before simpler tools

Packet Capture requires an agent, takes time to set up, and produces raw data that's hard to parse. Always try IP Flow Verify and Next Hop first — they answer 80% of questions in seconds.

Forgetting Connection Troubleshoot needs the NW agent

Unlike IP Flow Verify and Next Hop, Connection Troubleshoot requires the Network Watcher Agent VM extension installed and running on the source VM.

Enabling flow logs but not linking Traffic Analytics

Raw flow logs in storage are nearly unusable at scale. Always pair them with Traffic Analytics and a Log Analytics workspace for actionable visibility.

Not checking Effective Security Rules before running IP Flow Verify

Effective Security Rules shows all merged rules at a glance — often faster than running IP Flow Verify for multiple test cases to find the blocking rule.

Assuming IP Flow Verify tests the full path

IP Flow Verify only checks NSG rules — it doesn't test routing, DNS, or application-layer issues. A packet can be "allowed" by NSGs and still fail to reach its destination due to a misconfigured route.

Confusing Connection Monitor with Connection Troubleshoot

Connection Troubleshoot is a one-time test. Connection Monitor is a continuous service that runs on a schedule and alerts on failures — use Monitor for ongoing reliability tracking, Troubleshoot for immediate diagnosis.